



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Technical Drawing with AutoCAD Electrical

TGS Ref No: TGS-2023037466

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 6.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	23 May 2023	First version	Tertiary Infotech Academy
2.0	17 Sept 2024	Second version	Tertiary Infotech Academy
3.0		New format	Tertiary Infotech Academy
4.0	17 Oct 2025	Update company name	Tertiary Infotech Academy
5.0		Aligned with Master Trainer Slides v5	Tertiary Infotech Academy
6.0	18 August 2026	Full regeneration: single-source rebuild aligned to CP LU1–LU3; 8 hands-on labs with DWG starter files; schedule rebalanced across 2 days	Dr Alfred Ang

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Course Information

Course Title	Technical Drawing with AutoCAD Electrical
WSQ Course Reference	TGS-2023037466
TSC	Technical Drawing (DSN-TDR-4005-1.1)
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 16 hours (14 hours training + 2 hours assessment)
Daily Timing	9:30 am – 6:30 pm (45-min lunch; tea breaks within training time)
Mode	Instructor-led classroom with hands-on AutoCAD Electrical practical work
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Devise metrics for technical drawings production using AutoCAD Electrical and formulate solutions for encountered development problems. (K1, K2, A1, A2)
- LO2: Analyse technical drawings with AutoCAD Electrical to determine construction needs while reviewing for accuracy and adherence to specifications. (K3, K4, A3, A4)
- LO3: Review technical drawings for relevance to organisation's strategies and ensure their conformance to submission standards. (K5, K6, A5, A6)

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. (K1–K6)
- Practical Performance (PP) — hands-on AutoCAD Electrical tasks, 1 hour, open book. (A1–A6)
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Getting started, projects and schematic drawing

Time	Duration	Topic / Activity
0930–0945 hrs	15 min	Digital Attendance (AM) — SSG QR code. Welcome and course introduction (Slides 1–3)
0945–1015 hrs	30 min	Trainer and learner introductions, ground rules, course outline, learning outcomes, Skills Framework

		TSC (Technical Drawing, DSN-TDR-4005-1.1) and briefing for assessment (Slides 4–18)
1015–1055 hrs	40 min	Topic 1 — Introduction to Technical Drawing Using AutoCAD Electrical: user interface, symbols/standards/conventions (K1), ribbon, marking menu, palettes, status bar, model space, navigation (K2). Lecture, didactic questioning and demonstration (Slides 19–33)
1055–1105 hrs	10 min	Morning tea break
1105–1200 hrs	55 min	Topic 1 practical: Lab 1: Set Up a New AutoCAD Electrical Drawing (A1). Devise metrics: template, workspace, limits, units, grid and snap (Slides 34–43)
1200–1245 hrs	45 min	Lunch break
1245–1300 hrs	15 min	Digital Attendance (PM). Morning recap
1300–1420 hrs	80 min	Topic 1 — projects and drawing management: Project Manager, project/drawing properties, reference systems, title blocks (A1, A2). Hands-on: Lab 2: Create a Project with Project Manager (Slides 27–48)
1420–1450 hrs	30 min	Topic 2 — Analysing Technical Drawings: types of technical drawings (K3), technical drawing standards (K4), symbol library standards IEC/IEEE/NFPA (K1) (Slides 50–54)
1450–1500 hrs	10 min	Afternoon tea break
1500–1650 hrs	110 min	Topic 2 — schematic components, catalog browser, icon menu, parent/child, PLC modules, Circuit Builder (A4). Hands-on: Lab 3: Build Circuits with Circuit Builder (Slides 55–77)
1650–1815 hrs	85 min	Topic 2 — wires, wire layers, wire numbers, terminals (K4, A3). Hands-on: Lab 4: Insert Wires, Wire Numbers and Terminals (Slides 62–81)
1815–1830 hrs	15 min	Day 1 recap and Q&A

Total training/assessment time: 495 minutes (8.25 hours), excluding lunch.

Day 2 — Analysis, documentation, submission and assessment

Time	Duration	Topic / Activity
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0930–0945 hrs	15 min	Digital Attendance (AM). Day 1 recap
0945–1055 hrs	70 min	Topic 2 (continued) — signal arrows, cross-referencing, Surfer, Modify commands, project-wide updates, Mark/Verify (A3, A4). Hands-on: Lab 5: Cross-Reference and Surf a Drawing Set (Slides 66–86)
1055–1105 hrs	10 min	Morning tea break
1105–1200 hrs	55 min	Topic 3 — Reviewing Technical Drawings for Strategic Relevance and Standard Compliance: fundamentals of drafting design documentation (K5), panel layouts, nameplates, item numbers (Slides 89–104)
1200–1245 hrs	45 min	Lunch break
1245–1300 hrs	15 min	Digital Attendance (PM)
1300–1420 hrs	80 min	Topic 3 — panel layouts and documentation (A5). Hands-on: Lab 6: Create a Panel Layout with Footprints. Reports and part lists. Hands-on: Lab 7: Generate Reports and Part Lists (Slides 105–112)
1420–1450 hrs	30 min	Topic 3 — custom symbols with Symbol Builder (K6); external references; plot and publish to submission standards (A6). Hands-on: Lab 8: Plot and Publish the Project to PDF (Slides 99–116)
1450–1500 hrs	10 min	Afternoon tea break
1500–1550 hrs	50 min	Recap of all content; practice exam and revision for assessment (Slides 117–120)
1550–1630 hrs	40 min	Course feedback and TRAQOM survey. Briefing for assessment. Digital Attendance (Assessment) (Slides 121–126)
1630–1730 hrs	60 min	Final Assessment: Written Assessment — Short-Answer Questions (WA-SAQ), 1 hour, open book (K1–K6)
1730–1830 hrs	60 min	Final Assessment: Practical Performance (PP), 1 hour, open book (A1–A6). End of class

Total training/assessment time: 495 minutes (8.25 hours), excluding lunch.

Lab Reference (aligned to Learning Outcomes)

Topic / Learning Unit	Maps to	Labs
Topic 01: Introduction to Technical Drawing Using AutoCAD Electrical	LO1	Lab 1: Set Up a New AutoCAD Electrical Drawing; Lab 2: Create a Project with Project Manager
Topic 02: Analysing Technical Drawings	LO2	Lab 3: Build Circuits with Circuit Builder; Lab 4: Insert Wires, Wire Numbers and Terminals; Lab 5: Cross-Reference and Surf a Drawing Set
Topic 03: Reviewing Technical Drawings for Strategic Relevance and Standard Compliance	LO3	Lab 6: Create a Panel Layout with Footprints; Lab 7: Generate Reports and Part Lists; Lab 8: Plot and Publish the Project to PDF